HH–T–791c
April 14, 1965
SUPERSEDING
Fed. Spec. HH–T–791b
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FEDERAL SPECIFICATION

TUBE; FLEXIBLE, NONMETALLIC, ELECTRICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers nonmetallic, flexible tubes, hereinafter referred to as tubing. This specification does not cover electrical insulation sleeving.

1.2 Classification.

1.2.1 *Sizes.* The tubing shall be furnished in the following sizes as specified (see 6.2).

7/32 inch	5/8 inch	1-1/2 inches
1/4 inch	3/4 inch	2 inches
3/8 inch	1 inch	2-1/4 inches
1/2 inch	1-1/4 inches	2-1/2 inches

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

UU–T–111—Tape, Paper, Gummed (Sealing and Securing).

PPP–B–636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–P–116—Preservation, Methods of.

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Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, shall apply.

Underwriters' Laboratories (UL), Inc. Standard:

UL–3—Flexible Nonmetallic Tubing.

(Copies may be obtained from Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, N. Y., 11749; 207 E. Ohio Street, Chicago, Illinois, 60611; and 1655 Scott Blvd. Santa Clara, California, 95050.)

3. REQUIREMENTS

3.1 Material. The material used in the fabrication of the tubing shall meet the performance requirements specified herein, which shall be in conformance with the Underwriters' Laboratory Standard for Flexible Nonmetallic Tubing.

3.2 Smoothness of interior. Each length of flexible tubing shall be free from obstructions which could cause abrasion of the wire insulation and interfere with the insertion of an insulated wire of the maximum size used with the tubing.

3.3 Double-wall tubing. The construction of double-wall flexible tubing shall be such that the interior wall shall not be removable in lengths of more than 3 feet.

3.4 Size. Flexible tubing shall be fabricated in the sizes specified in 1.2.1. The internal diameter shall be not less than the specified size, except that the 7/32 inch size shall have an internal diameter of not less than 0.285 inch and the ¼ inch size shall have an internal diameter of not less than 0.320 inch.

3.5 Elongation. The elongation of the tubing shall not exceed 15 percent of its original size and show no evidence of breaking or opening at any point when tested as specified in 4.4.5.

3.6 Moisture absorption. The tubing shall not absorb more than 10 percent of its weight of moisture when immersed in water at 18° ± 3° centigrade (C.) for a period of 24 hours.

3.7 Flame resistance. When exposed to five (5) fifteen (15) second applications of a standard test flame, the tubing shall not convey flame nor continue to burn for more than one minute.

3.8 Kink and flattening resistance. The tubing shall not crack open, permanently kink or flatten after being tested as specified in 4.4.8. At the completion of this test the tubing shall easily permit the insertion of an insulated wire of the maximum size used with the tubing.

3.9 Flexibility. The 7/32, ¼, and ⅜ inch sizes of tubing, containing an insulated wire of the maximum size used with the tubing, shall be capable of being bent to a radius of ½ inch without injury to the tubing or coating.

3.10 Dripping of compound. The compound shall show no evidences of dripping when the flexible tubing is maintained at a temperature of 71° C. for a period of 15 minutes.

3.11 Identification. Flexible tubing shall have the manufacturer's trademark or a

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distinctive mark throughout its entire length by which it may be identified as a product of a particular manufacturer.

3.11.1 *Coil identification.* A coil of flexible tubing shall be marked or tagged to indicate the following:

(a) The manufacturer's name or trade name of the flexible tubing.

(b) The trade size of the tubing.

(c) The date of manufacture by month and year.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection provisions.

4.2.1 *Inspection lot.* Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL–STD–105.

4.2.2 *Sampling.* For the purpose of sampling, each coil of tubing shall be considered a unit of product.

4.2.3 *Examination and tests.* Examination and tests related to section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with MIL–STD–105, and the inspection level and sampling plans specified in table I. Examination and tests for packaging, packing, and marking shall be in accordance with MIL–P–116 and section 5 herein.

TABLE I.—*Classification of defects*

Use inspection level I of table I with sampling plan table II–A of MIL–STD–105

Critical:	*None*		
Major:	*AQL 1.0 percent defective*	*Requirement*	*Test procedure*
101.	Smoothness of material	3.2	4.4.2
102.	Size	3.4	4.4.4
103.	Coil identification	3.11.1	4.4.12
Minor:	*AQL 4.0 percent defective*		
201.	Identification	3.11	4.4.11

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4.2.4 *Disposition of nonconforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL–STD–105.

4.3 **Control tests.**

4.3.1 *Environmental and physical.* One coil of tubing shall be selected at random as control samples from each lot produced or from each month's production whichever occurs first. The control samples shall have successfully passed the tests specified in table I and shall then meet the following requirements and tests:

Control test	Requirements	Test procedures
301. Material	3.1	4.4.1
302. Double-wall tubing	3.3	4.4.3
303. Elongation	3.5	4.4.5
304. Kink and flattening resistance	3.8	4.4.8
305. Flexibility	3.9	4.4.9
306. Moisture absorption	3.6	4.4.6
307. Flame resistance	3.7	4.4.7
308. Dripping of compound	3.10	4.4.10

4.3.2 *Control test failures.* Should any one item of a control test sample fail to meet the specified test requirements acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 **Test methods and procedures.**

4.4.1 *Material.* The materials used in manufacture of the tubing shall be examined to insure compliance with 3.1. In addition to the tests specified herein, data shall be available showing evidence of material analysis prior to acceptance by certification.

4.4.2 *Smoothness of interior.* The interior surfaces of the tubing shall be visually examined for obstruction or damage by looking through a sample piece of tubing 8 inches in length aimed at a 75-watt, frosted, incandescent electric lamp to determine compliance with 3.2. In addition the 7/32, 1/4 and 3/8 inch sizes of tubing shall also be tested by inserting a length of standard single-braid, solid conductor insulated wire into a 20-foot length of the specified tubing. No. 14 AWG wire shall be used with 7/32 inch tubing; No. 10 AWG wire with 1/4 inch tubing; and No. 8 AWG wire with 3/8 inch tubing to determine compliance with 3.2.

4.4.3 *Double-wall tubing.* This test shall be conducted with a sample length of tubing 40 inches long. Expose the inner wall of the sample at one end for a length of two inches. With the exposed end secured in a vise, manually exert a strong steady pull on the opposite end of the sample to determine compliance with the requirements of 3.3.

4.4.4 *Size.* Measurement of the tubing sizes specified in 1.2.1, and their internal diameters as indicated by commercial trade sizes or as specified in 3.4 shall be made on 2 sample pieces of tubing 8 inches in length, with standard measuring equipment. The measured diameter of the tubing shall be equal to the minimum internal diameter permitted for the designated tubing size in compliance with the requirements of 3.4.

4.4.5 *Elongation.* Elongation of the tubing shall be determined as the permanent increase in length. Three sample pieces of tubing, 30 inches in length shall be prepared for this test by placing a mark at

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each extremity of the center 20-inch portion of the sample. Reinforce each end of a sample with a length of wood or steel 5 inches in length and fasten clamps to these ends so that the ends project 2 inches beyond the edge of the clamps. Attach a weight of 150 pounds to one clamp and secure the other clamp to a pulley system so that the sample hangs vertically for its full length at right angles to the faces of the clamps. The sample, clamps, and weight shall then be raised gradually so that the weight clears the ground and hangs freely in the air, and the weight supported by the sample for 5 minutes.

4.4.6 *Moisture absorption.* Three samples, 8 inches in length, removed from the same coil of finished tubing shall be cut with square ends and prepared for this test by closing each end with a cork, without distortion of the material, and then completely sealing the ends with paraffin. The samples shall be weighed on scales and the weight recorded to the nearest 0.5 gram. The samples shall then be completely immersed for 24 hours in water maintained at a temperature of 18° ± 3°C. At the end of this immersion period the samples shall be removed from the water. The surface water shall be quickly absorbed by blotting the sample lightly with dry cloth or blotting paper and then each sample shall be cut open to allow any water on the inside to drain out. The samples shall then be reweighed on the same scales within 3 minutes of removal from the water and the increase in weight shall not be more than 10 percent of the previous recorded weights in compliance with the requirements of 3.6.

4.4.7 *Flame resistance.* This test shall be conducted under draft free conditions. Three sample pieces of tubing 18 inches in length shall be prepared for this test by closing the upper ends to prevent updraft through the tubing. The sample shall be secured in a sheet metal enclosure 12 inches wide, 14 inches deep, 24 inches high and open at the top with means provided for supporting the sample in a vertical position and keeping it taut during the test. Attach a ½ inch strip of gummed paper tape, class 2 of UU–T–111, as a flame indicator, around the sample so that the lower edge is 10 inches above the point at which the inner blue flame of the test flame is to be applied. The paper flame indicator shall be wrapped once around the tubing with the ends pasted evenly together and projecting ¾ inch from the tubing on the opposite side of the sample to which the flame is to be applied. The height of the flame with the burner vertical shall be adjusted to 5 inches, with an inner cone 1-½ inches high. The burner shall be positioned with only the pilot lighted in front of the tubing, on a 20-degree angle so that the vertical plane through the stem of the burner includes the axis of the tubing with the distance 1-½ inches along the extended axis of the burner stem, between the tip of the stem and the surface of the tubing. The sample shall be subjected to 5 complete cycles of application of the flame for 15 seconds and removal for 15 seconds. There shall be not more than 25 percent of the paper flame indicator burned after the 5 applications of the flame and the duration of burning of the sample after the fifth application of flame shall not exceed one minute in compliance with the requirements of 3.7.

4.4.8 *Kinks and flattening resistance.* The test for kink and flattening resistance shall be performed on 3 sample pieces of tubing 24 inches in length. Kinks shall be made in the sample by holding it tightly at its ends and manipulating it. The tubing shall be capable of having the kinks removed by pulling on the tubing. Flattening resistance shall be tested by flattening a 6-inch portion of the tubing so that the inner walls are forced together. The injured portion shall be manipulated by hand until the exterior is restored to roundness. The injured portion of the sample shall then be cut out and the interior visually examined by looking through with the aid of a 75-watt frosted, incandescent lamp.

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The tubing shall not crack open as a result of the flattening and the interior shall be restored to sufficient roundness and free from projecting edges or lumps which would prevent the insertion of the maximum size of insulated wire used with the tubing in compliance with 3.8.

4.4.9 *Flexibility.* Three sample pieces of tubing 8 inches in length shall be exposed to a controlled temperature of 22°C. for 15 minutes prior to test. While still exposed to this temperature, insulated wire of the maximum size used with the designated tubing shall be inserted into a sample and the sample wrapped completely around a one inch mandrel at a tension sufficient only to cause the tubing to conform closely to the periphery of the mandrel. Upon removal from the mandrel the sample shall be straightened and visually examined for injury to the tubing or coating in compliance with 3.9.

4.4.10 *Dripping of compound.* This test shall be performed on 2 samples of tubing 8 inches in length, in a chamber capable of maintaining a temperature of 71° ± 3°C. The sample shall be suspended vertically in the chamber which has been preheated to the required temperature. A sheet of paper or container shall be placed in the bottom of the chamber to collect any drippings from the sample. The sample shall be exposed to the 71° ± 3°C. temperature for 15 minutes. At the end of this exposure period the sample and the area underneath shall be visually examined for evidence of dripping to determine compliance with 3.10.

4.4.11 *Indentification.* The outer surface of the tubing shall be visually examined for presence of a distinctive mark which readily identifies the tubing as the product of a particular company. This identification shall be repeated at intervals throughout the entire length of tubing in compliance with 3.11.

4.4.12 *Coil identification.* The coil of tubing shall be visually examined to determine compliance with the requirements specified in 3.11.1.

5. PREPARATION FOR DELIVERY

(For civil agency procurement the levels of preservation, packaging, and packing used are defined in Fed. Std. No. 102.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2.)

5.1.1 *Level A.*

5.1.1.1 *Cleaning.* The tubing shall be cleaned to remove any deposit foreign to the composition of the item in accordance with process C–1 of MIL–P–116.

5.1.1.2 *Unit packaging.* The tubing shall be coiled and secured with at least 5 ties of ¼-inch wide flat cotton tape and packaged in a close-fitting fiberboard box conforming to W6c of PPP–B–636. Closure shall be in accordance with the appendix of the box specification.

5.1.2 *Level C.* The tubing shall be packaged in accordance with the manufacturer's commercial practice which shall afford adequate protection against deterioration and physical damage from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.* Identical coils of tubing packaged as specified in 5.1 shall be packed in a fiberboard box conforming to V3c of PPP–B–636. Closure and strapping shall be as specified in the appendix of the box specification.

5.2.2 *Level B.* Identical coils of tubing packaged as specified in 5.1 shall be packed in a fiberboard box conforming to W5c of PPP–B–636. Closure and strapping shall

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be as specified in the appendix of the box specification.

5.2.3 *Level C.* Coils of tubing shall be packed in a manner that will provide protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

5.3 Marking.

5.3.1 *Military.* Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 **Intended use.** Tubing covered by this specification is used as a mechanical protection over insulated wires and cables.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.

(*b*) Size of tubing required (1.2.1).

(*c*) Inspection responsibility, if other than specified (4.1).

(*d*) Inspection lot size, etc., if other than specified (4.2.1).

(*e*) Level of packaging and packing required (5.1 and 5.2).

(*f*) Special marking, if required (5.3.2).

6.3 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Conduits, flexible, other than steel, for other than underground work.

Carload minimum weight 30,000 pounds.

Motors:

Conduits, for other than underground work, flexible, other than steel.

Truckload minimum weight 30,000 pounds subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—MU

Navy—YD

Air Force—85

Review activities:

Army—MU, EL, MO

Navy—SH

Air Force—85

User activities:

Army—MI, WC

Navy—YD, MC

Air Force—None

Civil Interest:

General Services Administration

Preparing activity:

Army—MU

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